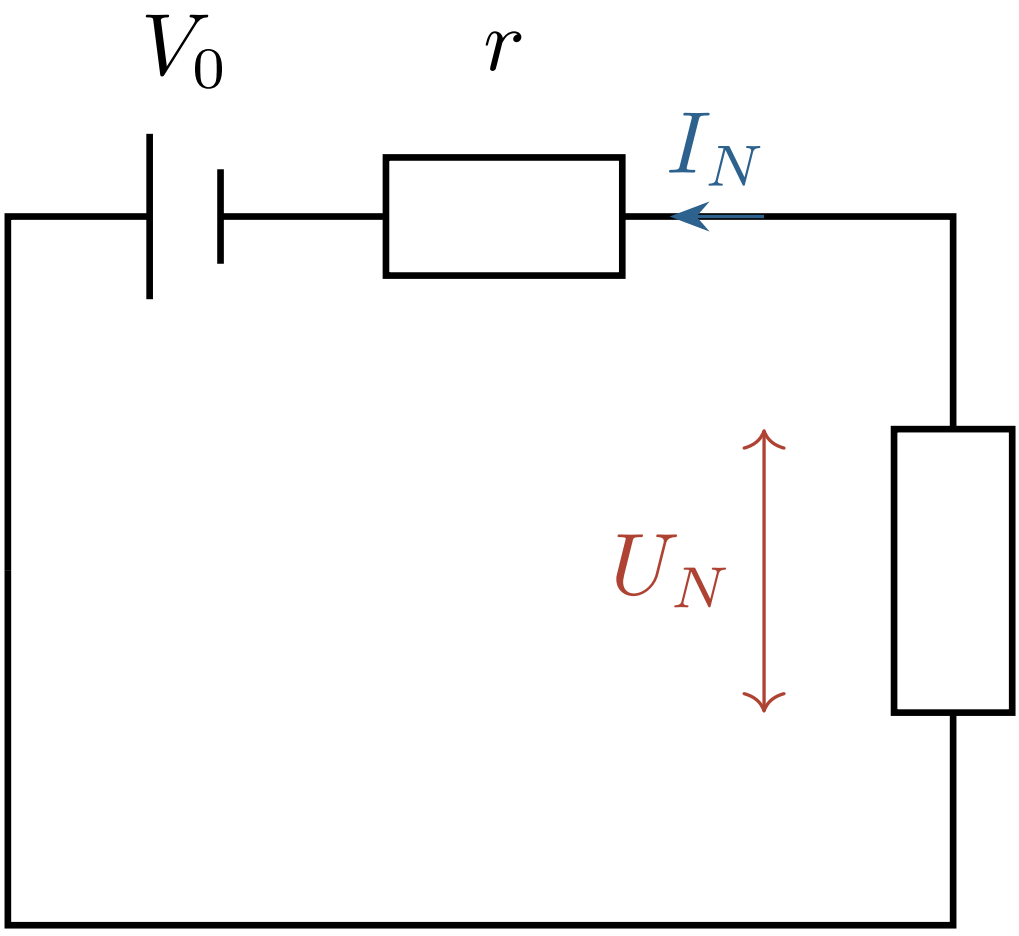

$$V_0 I_N = P_N + I_N^2 r$$

$$P_N = I_N^2 R_\mathrm{L}$$